

[Collaborative Docs](#) >

Discussion Agenda 08/23/2015

Below is a discussion agenda / live progress report in an approximate order of importance for the brainstorming session of the 4th AGORA Workshop @ University of California Santa Cruz (Aug. 21–23, 2015; see the [program](#)).

Per Steering Committee's recommendation, Paper 4 ([workspace](#)) will be an area of focused AGORA efforts before and during this year's Workshop.

Contents

- [1 1. Progress Report \(as of Aug. 2015\)](#)
- [2 2. Paper Group 4 – Progress and Discussion](#)
- [3 3. Other Topics for Collaboration-wide Discussion](#)
- [4 4. Recaps](#)
- [5 5. Your Feedback](#)

[1. Progress Report \(as of Aug. 2015\)](#)

We are actively making a lot of progress, chiefly, in three different fronts below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																																																																
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo)	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Calibrate SF/Fbck recipes using common disk ICs																																																																
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																																
PARTICIPANTS	• Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) • Code representatives:	• Working Groups engaged: - I (common physics) - III (common cosmo ICs) • Code representatives:	• Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) • Code representatives:																																																																
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			Thompson RAMSES Agertz, Butler, DeGraf, Teyssier, Nickerson (Paper 4 code leaders, as of August 2015)
PROGRESS MADE OR BEING MADE	Please visit Paper 2's Workspace where Oliver Hahn et al. have been updating the progress report.	Either via Britton's original interface or through Nick's API, GRACKLE is now fully implemented (equilibrium cooling version) in all 9 participating codes. The resulting cooling functions is being compared and verified in isolated disk runs (Paper "4").	- Please visit Paper 4's Workspace where Oscar Agertz et al. have been updating the progress report. - All 9 participating codes have submitted their "first" datasets (i.e. Grackle + noSF runs) as of August 2015. First composite figures have been posted on Aug. 9, 2015. Please visit Paper 4's Workspace for much detailed, up-to-date information. - The agora-analysis-script repository is being updated as the analysis progresses. - Meanwhile, yt-3.2 is released on July 24, 2015 thanks in part to many AGORA members including Kenza Arraki, Ben Keller, Britton Smith, and Matt Turk. See the announcement. This or even more recent yt-dev version will be needed for the aforementioned agora-analysis-script to properly work on isolated disk datasets.
OTHER LINKS	- Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. MUSIC Google group	GRACKLE Google group GRACKLE download & documentation AGORA API for GRACKLE by Nick Gnedin	- Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. yt-AGORA Google group

[2. Paper Group 4 - Progress and Discussion](#)

Below are abbreviated talking points from the [Paper 4 collaborative doc](#) only for this telecon. The writings below will not be updated as the ideas in Paper "4" evolve in time. **Participants of Paper "4" are STRONGLY recommended to check out and follow the up-to-date consensus found in the**

Paper 4 collaborative doc.

(1) "First" Datasets (Grackle+noSF)

- In addition to the consensus for the "first" dataset made previously,
 - AMR groups will redo their runs with: (a) `vcirc_SPH.dat`, (b) common refinement criteria (refine once a gas cell contains 1–8x the SPH gas particle mass or 8 collisionless -- DM or IC stars -- particles).
 - SPH groups will redo their runs with: (a) pressure support with $\Delta x = 80$ pc, (b) minimum smoothing length = $0.2 \times \text{gravitational softening length}$.
- Make sure to have an initial metallicity = 0.02041 (= 1 Z_{sun} in Grackle v2.0, but = 1.5761 Z_{sun} in Grackle v2.1).
- Resolution comparison by a subset of groups: find an optimal combination of refinement parameters to make AMR compatible with SPH (Keller & Feldmann volunteered).
- We will eventually use median-resolution runs (8.593e3 M_{sun} or 40 pc) for the Paper: stellar feedback schemes to be used in cosmological runs are calibrated at this resolution.

(2) "Second" Datasets (Grackle+SF+ThermalFbck)

- In addition to the consensus prescription for SF/ThermalFbck made previously, we further elaborate on:
 - (a) star formation prescription: minimum star particle mass = gas particle mass in SPH
 - (b) stellar feedback prescription: 1 SN explosion event per star particle, Per 91 $M_{\odot}$ of stellar mass 1 SN occurs, 2.63 $M_{\odot}$ of metals and 14.8 $M_{\odot}$ of gas (including metals) gets released, 5 Myr lifetime, deposit schemes left at participants' discretion

(3) "Third" Datasets (Grackle+SF+FavoriteFbck)

- Calibrate our feedback so it has an impact at galactic scale.
- Figure-of-merit to decide which galaxy looks reasonable?

3. Other Topics for Collaboration-wide Discussion

(1) Ratification of the AGORA SMBH Prescription (Accretion and Feedback):

- Visit [this link](#) for background information.
- Thoughts from Alex Hobbs and a consensus so far: Keep it as simple as possible, and perhaps later to allow groups to implement their own models/improvements as 2nd-order 'bonus' science. His Working Group's recommendations, to which we agreed, are the following:
 - (a) Bondi–Hoyle accretion model with alpha modification based on ratio of n_{H} (Booth & Schaye 2009), Eddington–limited
 - (b) Kernel-weighted thermal feedback, energy is released every timestep according to accretion rate (i.e. no 'bottled-up' feedback), we recommend mass-weighted energy deposition for AMR codes
 - (c) $1e9$ K temperature ceiling
- Still open to ideas for seeding/merging. Thoughts from Colin DeGraf, Lucio Mayer, and others?

(2) yt–Sunrise Pipeline for AGORA on NERSC:

- Visit [this link](#) for background information about yt–Sunrise pipeline ([preliminary documentation](#)) for AGORA on NERSC (Data Pilot Resources) developed by Miguel Rocha et al.
- Who will continue to work on this, and integrate it into a public yt? See its current status and loose ends [here](#). Also check out the [status](#) of Theia hardware volume rendering interface now removed from the public yt.
- Thoughts from Greg Snyder, Joel Primack, and others?
 - > Greg Snyder has volunteered to optimize the current pipeline and reincorporate back into yt–dev.
 - > Meanwhile, check out Ben Keller's contribution below if you'd like to try out Sunrise on your local resources, without having to install anything:
 - <https://www.physics.mcmaster.ca/~kellerbw/posts/sunrise-on-docker.html>
 - <https://hub.docker.com/r/bwkeller/sunrise/>

(3) AGORA Data Storage on NERSC or NCSA:

- We may choose to migrate from NERSC (LBNL/DOE-site) to NCSA (UIUC/NSF-site) as we may not expect GPU support for visualization at NERSC.
- Thoughts from Joel Primack? --> It is probably premature to move the pipeline/storage at this point.

(4) Working Group / Paper Group Leaderships, Future Timeline of the Project in General:

- Paper Group 2 (divide it into two?: identifying convex hull ICs + high-res DM-only sims)
--> Convex hull or ellipsoidal ICs for the $1e10$ and $1e13$ Msun halos exist. An ellipsoidal IC for the $1e11$ Msun quiescent halo exists. Christine Simpson has volunteered to obtain and share the convex hull ICs for the two $1e12$ Msun halos.
- Paper Group 3 (start right now?)
--> Because Paper "3" and "4" shares technologies and participants, maybe the best way to move forward is for the whole Paper Group "4" to naturally transition to Paper "3" as soon as Paper "4" is completed. Will consider using the $1e12$ Msun halos instead of the $1e11$ Msun halo.
- Working Group XIII (SMBH accretion and feedback)
- 1 or 2 science questions that we will try to sell to the potential funding agencies? (thus, 1–2 Science-oriented Working Groups that will be the focus in the next year)

(5) Next Source of Funding for AGORA:

- See the news release [here](#). Any suggestions?
- Thoughts from Joel Primack, Steering Committee members, and others?

4. Recaps

(1) Strategy to Store Simulation Outputs:

Please visit [this link](#).

(2) Mini-workshop Opportunities:

Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(3) Useful Links:

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)

5. Your Feedback

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!